

Week3-4_Assignment

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DSC 640: Data Presentation and Visualization 01/17/2026

Load Libraries

```
import os
import warnings
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib.ticker as mtk
import numpy as np
from tqdm import tqdm

In [70]:

# Set the directory
directory = r"C:\Users\velaz\OneDrive\Desktop\DSC_640"
os.chdir(directory)

warnings.filterwarnings('ignore')
```

Load Data

```
In [71]:

plt.style.use('default')
sns.set_palette('husl')
base_path = r"C:\Users\velaz\OneDrive\Desktop\DSC_640\Week - 3-4"

april = pd.read_csv(base_path + r'\2022.04_WAVES-ACCESS-RECORDS.csv')
may = pd.read_csv(base_path + r'\2022.05-WAVES-ACCESS-RECORDS.csv')
june = pd.read_csv(base_path + r'\2022.06_WAVES-ACCESS-RECORDS.csv')
july = pd.read_csv(base_path + r'\2022.07_WAVES-ACCESS-RECORDS.csv')
august = pd.read_csv(base_path + r'\2022.08_WAVES-ACCESS-RECORDS.csv')

print('Loaded datasets')

Loaded datasets
```

Data Cleaning, processing and engineering

```
In [72]:

# standardize column names across variants
def coalesce_columns(df, canonical, variants):
    if canonical in df.columns:
        return df
    for v in variants:
        if v in df.columns:
            df = df.rename(columns={v: canonical})
            break
    return df

frames = [april.copy(), may.copy(), june.copy(), july.copy(), august.copy()]

# Map of canonical to possible variants
col_variants = {
    'NAMELAST': ['Last Name'],
    'NAMEFIRST': ['First Name'],
    'NAMEMID': ['Middle Initial'],
    'ACCESS_TYPE': ['Access Type'],
    'APPT_MADE_DATE': ['Appointment Made Date'],
    'APPT_START_DATE': ['Appointment Start Date'],
    'APPT_END_DATE': ['Appointment End Date'],
    'APPT_CANCEL_DATE': ['Appointment Cancel Date'],
    'TOTAL_PEOPLE': ['Total People'],
    'LAST_UPDATEDBY': ['Last Updated By'],
    'LASTENTRYDATE': ['Last Entry Date'],
    'TERMINAL_SUFFIX': ['Terminal Suffix'],
    'VISITEE_NAMELAST': ['Visitee Last Name'],
    'VISITEE_NAMEFIRST': ['Visitee First Name'],
    'MEETING_LOC': ['Meeting Location'],
    'MEETING_ROOM': ['Meeting Room'],
    'CALLER_NAME_LAST': ['Caller Last Name'],
    'CALLER_NAME_FIRST': ['Caller First Name'],
    'CALLER_ROOM': ['Caller Room'],
    'RELEASEDATE': ['Release Date']
}

# Standardize each monthly dataframe
standardized = []
for i, df in enumerate(frames):
    for canonical, variants in col_variants.items():
        df = coalesce_columns(df, canonical, variants)
    standardized.append(df)

# Concatenate
wh = pd.concat(standardized, ignore_index=True)

# Parse date columns
date_cols = ['TOA', 'POA', 'TOD', 'POD', 'APPT_MADE_DATE', 'APPT_START_DATE', 'APPT_END_DATE', 'APPT_CANCEL_DATE',
              'LASTENTRYDATE', 'RELEASEDATE']
for c in date_cols:
    if c in wh.columns:
        wh[c] = pd.to_datetime(wh[c], errors='coerce', format='mixed')

# Feature engineering
wh['APPT_DATE'] = pd.to_datetime(wh['APPT_START_DATE'].dt.date, errors='coerce')
wh['APPT_MONTH'] = wh['APPT_DATE'].dt.to_period('M')
wh['APPT_MONTH_STR'] = wh['APPT_START_DATE'].dt.strftime('%Y-%m')

wh['VISITEE_FIRST'] = wh['VISITEE_NAMEFIRST'].astype(str) if 'VISITEE_NAMEFIRST' in wh.columns else ''
wh['VISITEE_LAST'] = wh['VISITEE_NAMELAST'].astype(str) if 'VISITEE_NAMELAST' in wh.columns else ''
wh['VISITEE_FULL'] = (wh['VISITEE_FIRST'].str.strip() + ' ' + wh['VISITEE_LAST'].str.strip()).str.strip().replace('', 'Unk')

wh['MEETING_LOC'] = wh['MEETING_LOC'].astype(str).fillna('Unknown').replace('', 'Unknown')
wh['MEETING_ROOM'] = wh['MEETING_ROOM'].astype(str).fillna('Unknown').replace('', 'Unknown')

if 'APPT_MADE_DATE' in wh.columns and 'APPT_START_DATE' in wh.columns:
    wh['LEAD_DAYS'] = (wh['APPT_START_DATE'] - wh['APPT_MADE_DATE']).dt.days
else:
    wh['LEAD_DAYS'] = np

wh['WEEKDAY'] = wh['APPT_DATE'].dt.day_name()

# Save consolidated dataset for safe keeping
output_dir = os.path.join(base_path, 'wh_outputs')
if not os.path.exists(output_dir):
    os.makedirs(output_dir, exist_ok=True)

consolidated_path = os.path.join(output_dir, 'wh_consolidated_apr_aug_2022.csv')
wh.to_csv(consolidated_path, index=False)

# Quick checks
print(wh.head())
print(wh[['APPT_DATE', 'APPT_MONTH_STR', 'ACCESS_TYPE', 'MEETING_LOC', 'MEETING_ROOM']].head())
print('Consolidated shape: ' + str(wh.shape))
print('Saved consolidated to: ' + consolidated_path)

NAMELAST NAMEFIRST NAMEMID UIN BDGNBR ACCESS_TYPE \
0 ABBOTT CHARLOTTE M U46318 NaN VA
1 ABBOTT CYNTHIA P U44131 183014.0 VA
2 ABBOTT ISAAC J U46474 NaN VA
3 ABBOTT KATIE A U46474 NaN VA
4 ABDELALL BRENDA F U46985 180837.0 VA

TOA POA TOD POD ... CALLER_ROOM RELEASEDATE APPT_DATE \
0 2022-04-23 07:49:00 NaT NaT NaT ... NaN 2022-08-02 2022-04-23
1 2022-04-27 09:55:00 NaT NaT NaT ... NaN 2022-08-02 2022-04-07
2 2022-04-29 10:29:00 NaT NaT NaT ... NaN 2022-08-02 2022-04-29
3 2022-04-29 10:31:00 NaT NaT NaT ... NaN 2022-08-02 2022-04-29
4 2022-04-26 13:17:00 NaT NaT NaT ... NaN 2022-08-02 2022-04-26

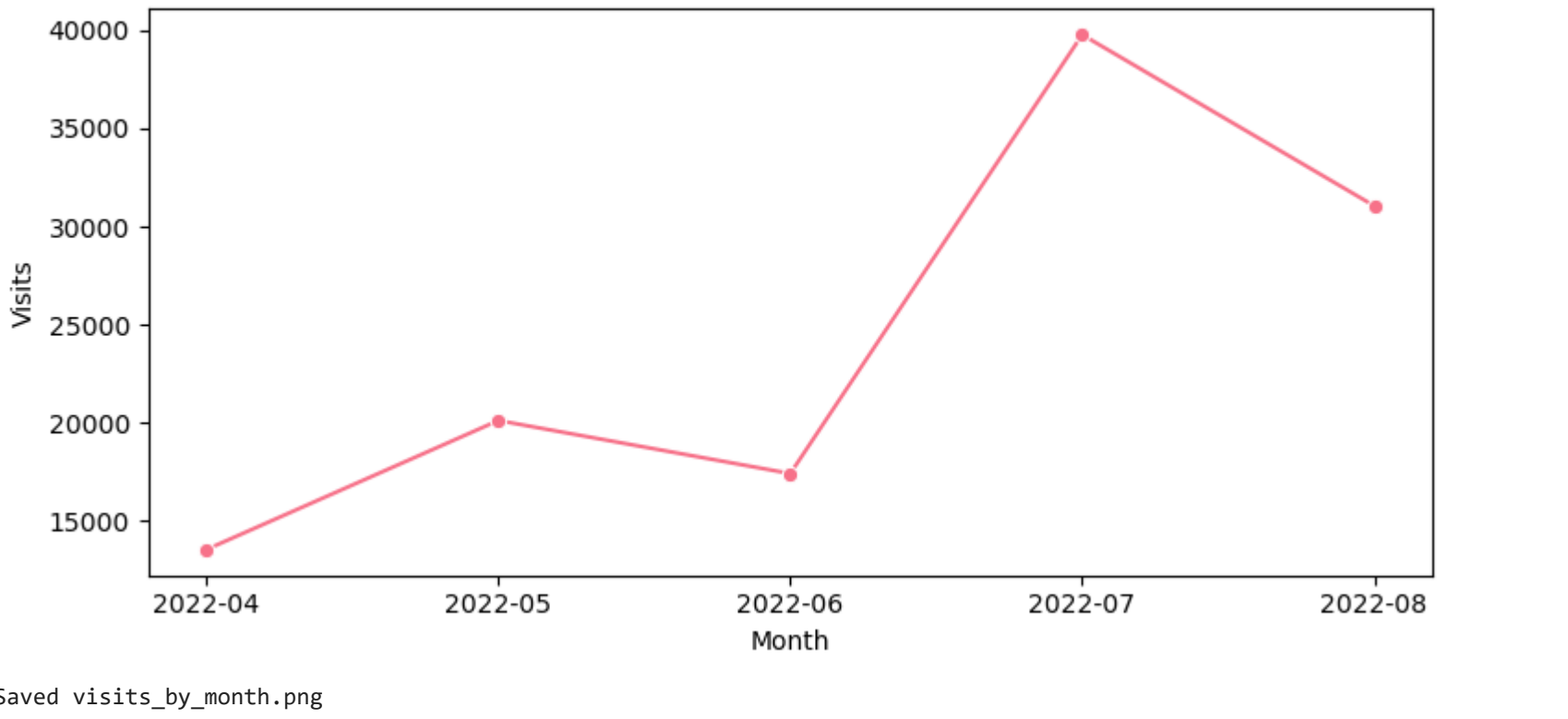
APPT_MONTH APPT_MONTH_STR VISITEE_FIRST VISITEE_LAST VISITEE_FULL \
0 2022-04 2022-04 Visitors Office Visitors Office
1 2022-04 2022-04 Erika Poethig Erika Poethig
2 2022-04 2022-04 Visitors Office Visitors Office
3 2022-04 2022-04 Visitors Office Visitors Office
4 2022-04 2022-04 William McIntee William McIntee

LEAD_DAYS WEEKDAY
0 1.0 Saturday
1 0.0 Thursday
2 6.0 Friday
3 6.0 Friday
4 0.0 Tuesday

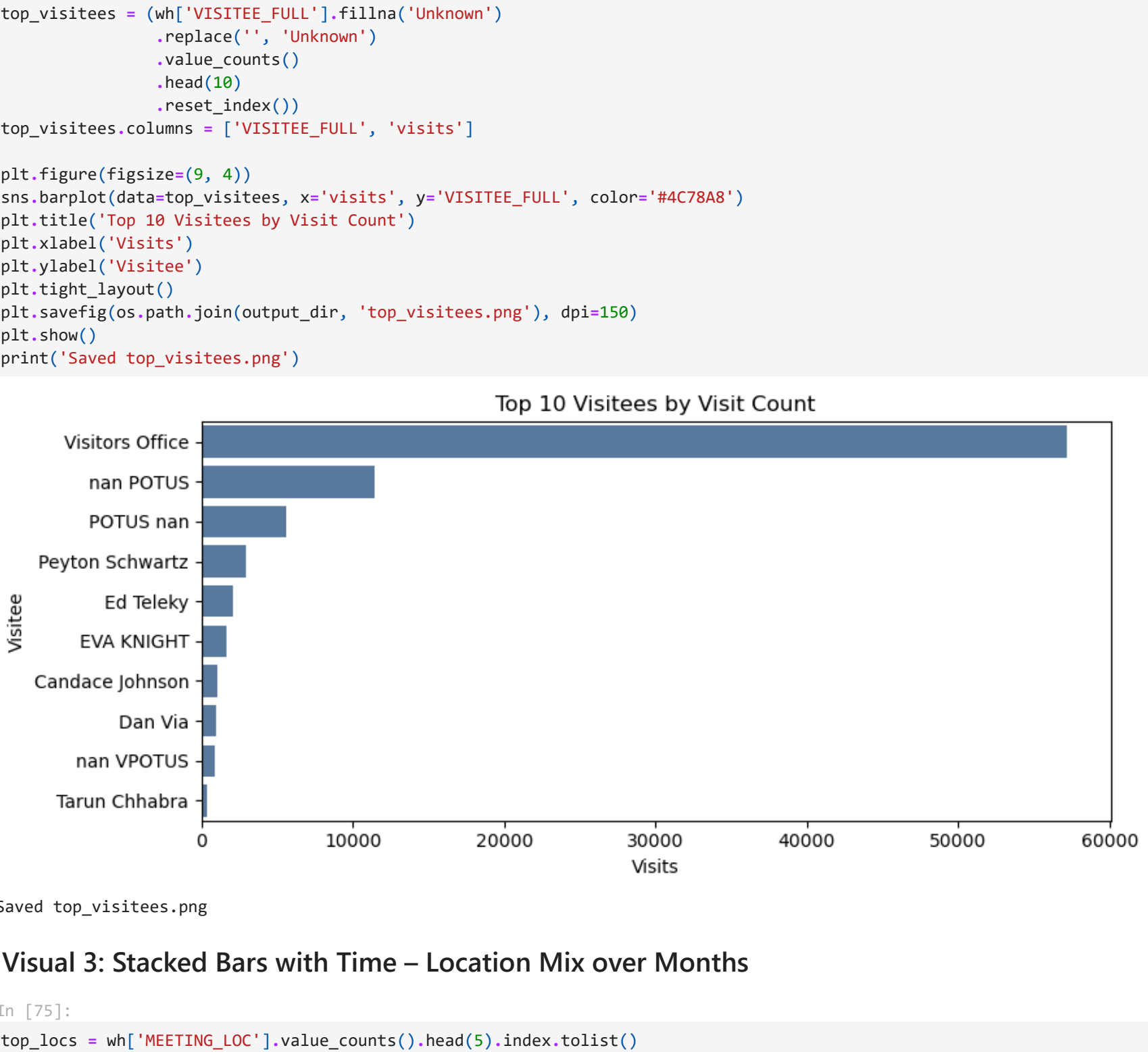
[5 rows x 35 columns]
APPT_DATE APPT_MONTH_STR ACCESS_TYPE MEETING_LOC MEETING_ROOM
0 2022-04-23 2022-04 VA WH EW TOUR
1 2022-04-07 2022-04 VA OE0B 167
2 2022-04-29 2022-04 VA WH EW TOUR
3 2022-04-29 2022-04 VA WH EW TOUR
4 2022-04-26 2022-04 VA OE0B 210/212 Di
Consolidated shape: (121864, 35)
Saved consolidated to: C:\Users\velaz\OneDrive\Desktop\DSC_640\Week - 3-4\wh_outputs\wh_consolidated_apr_aug_2022.csv
```

Visuals

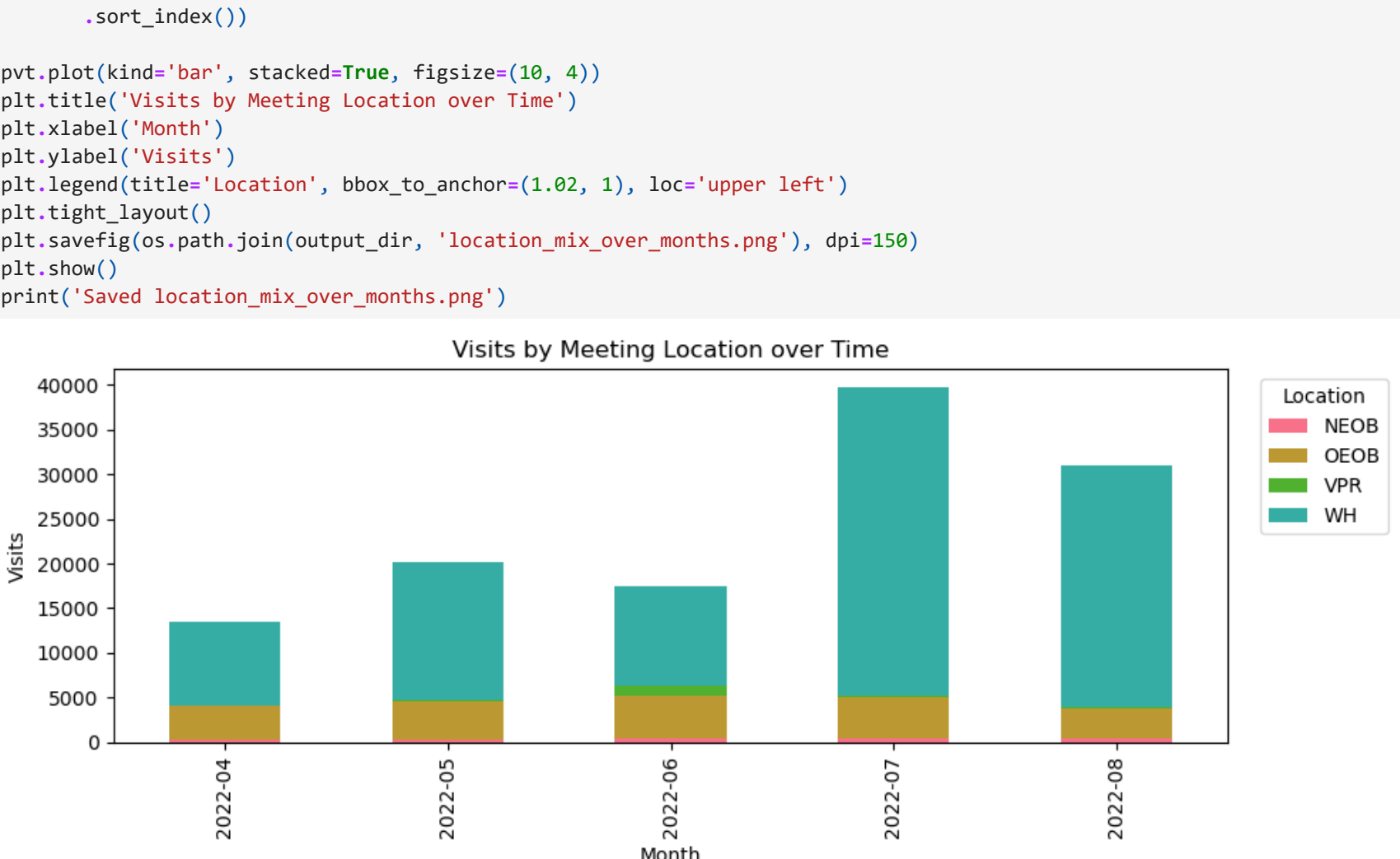
Visual #1: Line Chart



Visual 2: Bar Chart – Top 10 Visitees



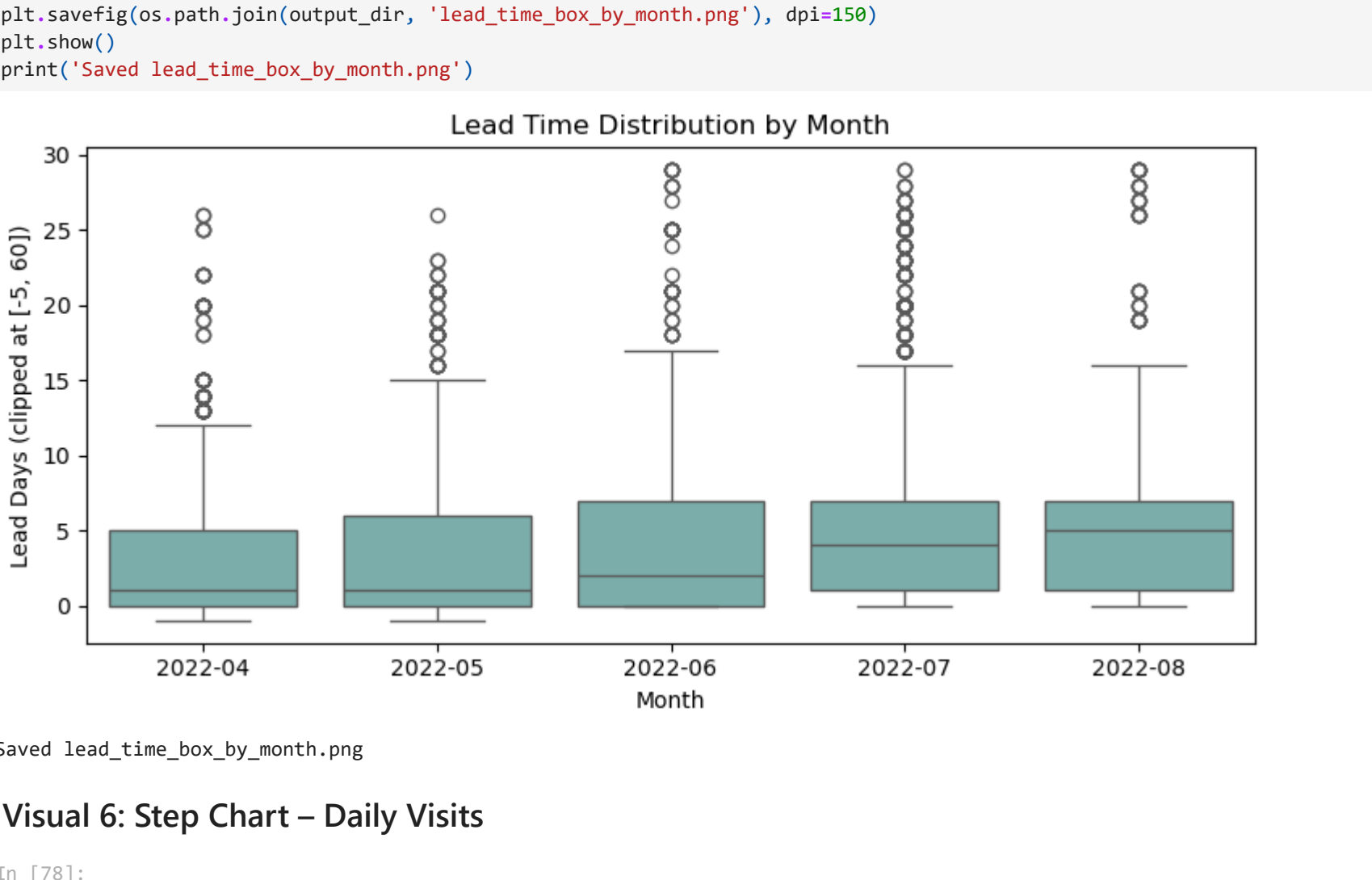
Visual 3: Stacked Bars with Time – Location Mix over Months



Visual 4: Heatmap



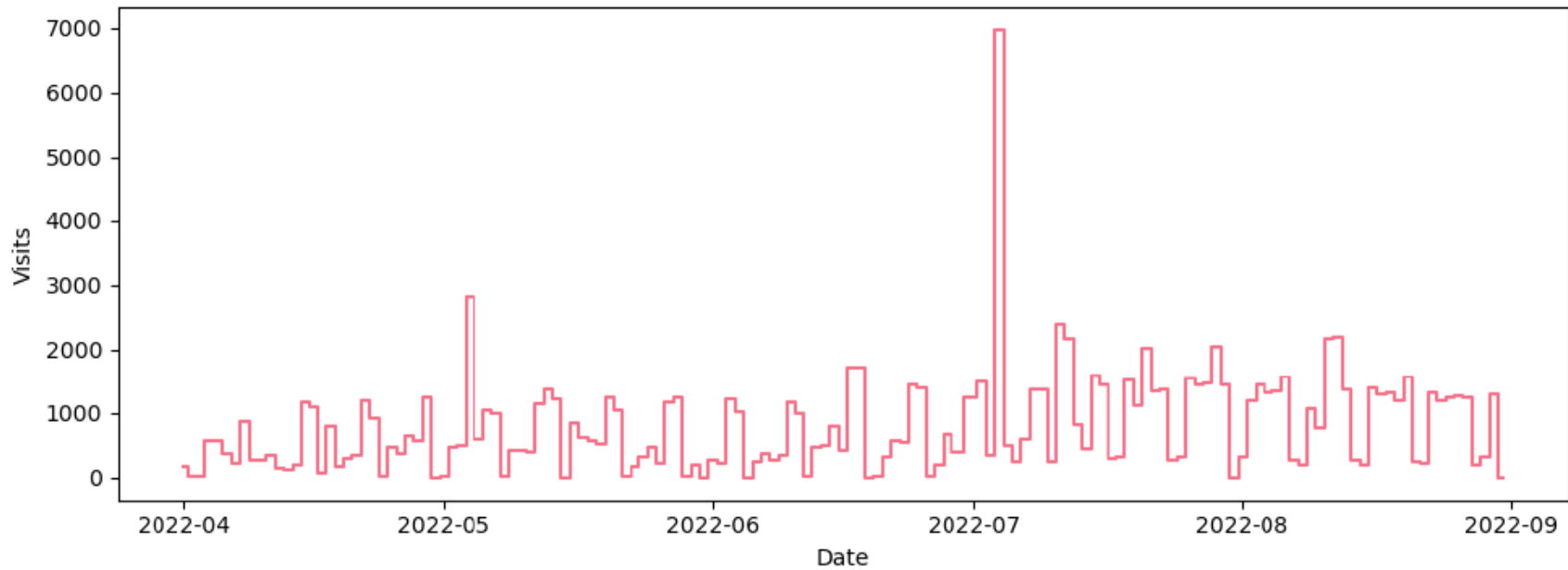
Visual 5: Box Plot – Lead Time Distribution by Month



Visual 6: Step Chart – Daily Visits



Daily Visits (Step)



Saved daily_step_visits.png

White House Visitor Activity Brief

(Apr-Aug 2022)

Audience:

Senior operations staff and communications leads at the Executive Office with intermediate data literacy.

Purpose and Call to Action:

Align staffing and scheduling with observed demand patterns to improve visitor experience and operational predictability. Approve a pilot program that (1) shifts large group tours away from mid-week peaks, (2) allocates additional screening and guidance staff on historically busy weekdays, and (3) sets a minimum seven day lead time for large group requests to reduce day of volatility.

Medium: A concise six chart dashboard shared in a brief meeting and as a link for weekly review. The format accelerates decisions while allowing asynchronous follow up.

Design:

We applied Gestalt principles for clarity and speed of comprehension: similarity (consistent colors for locations and months), proximity (co locating related metrics), continuity (left to right time flows), closure (titles that state the takeaway), and figure ground (ample white space and restrained palette). Axes, labels, and ordering are standardized across visuals to lower cognitive load.

Visuals and Insights: -

- Line chart (Visits by Month): Volume remains strong from April through August, indicating sustained public and internal engagement that requires steady capacity planning.
- Bar chart (Top Visitors): The Visitors Office and a small number of principals account for a large share of traffic, suggesting outsized operational leverage from focused scheduling with these offices.
- Stacked bar with time (Location Mix over Months): The mix across WH, OEOB, and Other sites shows distinct operational footprints; coordinating queues and security at OEOB merits special attention during peaks.

- Step chart (Daily Visits): A clear mid week crest indicates when to schedule more screening and wayfinding staff. Weekend dips suggest opportunities to shift tour blocks off peak weekdays.
- Heat map (Visits by Day and Hour): Concentrations cluster in business hours, with sharper intensity mid week; staggered entry windows could smooth short lived bottlenecks.
- Box plot (Lead Time by Month): Wide variability and outliers reveal operational risk from short notice bookings. A seven day lead time guideline for large groups would stabilize prep and reduce last minute conflicts.

Ethical Considerations:

We used public WAVES records and kept analyses at aggregated levels without deriving new PII. Transformations included date parsing, grouping long tail locations as Other, and aggregating by day and month. No records were removed; missing values are labeled Unknown. Risks include misinterpreting grouped categories and assuming APPT START DATE approximates actual arrival; mitigations include clear legends, explicit notes on grouping rules, and periodic validation against operations logs. We recommend documenting any policy changes and monitoring impacts on throughput, wait times, and visitor satisfaction to ensure equitable access.